UG10204

K32W1 / MCXW71 / MCXW72 Connectivity Test Application Getting Started User Guide

Rev. 1.0 — 24 January 2025 | User guide

Document information

Information	Content
Keywords	UG10204, K32W1, MCW71, and MCXW72 family of wireless microcontrollers, Connectivity Test Application, K32W148-EVK, FRDM-MCXW71, FRDM-MCXW72, MCX-W71-EVK, and MCX-W72-EVK hardware platforms, binary file, IAR IDE workspace file, flashing and running the application
Abstract	This document describes briefly how to build the Connectivity Test Application for the Arm Cortex based families of wireless microcontroller platforms provided by NXP. The application supports the NXP hardware platforms: K32W148-EVK, FRDM-MCXW71, FRDM-MCXW72, MCX-W71-EVK, and MCX-W72-EVK.

1 Introduction

This document describes briefly how to build the Connectivity Test Application for the Arm Cortex based K32W1, MCXW71, and MCXW72 wireless microcontroller platforms provided by NXP.

2 Building the binary file

This section describes the required steps for obtaining the binary file for usage with the board. The Connectivity Test Application is available in the package location below:

```
<sdk_folder_path>\boards\<board_name>\wireless_examples
\ieee-802.15.4\connectivity_test\bm\iar
```

Where <board name> is:

- k32w148evk for K32W1480-EVK
- frdmmcxw71 for FRDM-MCXW71
- frdmmcxw72 for FRDM-MCXW72
- mcxw71evk for MCX-W71-EVK
- mcxw72evk for MCX-W72-EVK

3 Building and flashing the Connectivity Test application using IAR

1. **Step 1**: Navigate to the Connectivity Test Application location described above.
2. **Step 2**: Open the highlighted IAR IDE Workspace file (*.eww file format:)

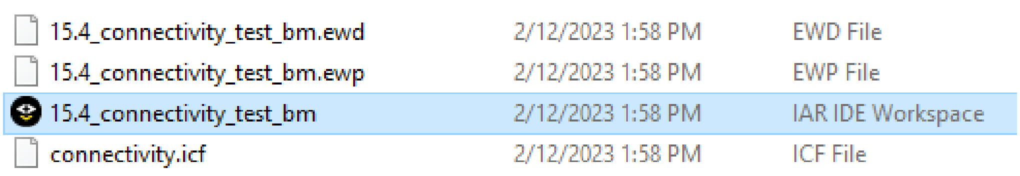

Figure 1. Connectivity Test Application Workspace file

3. **Step 3**:Select the desired configuration for the Connectivity Test Application project:

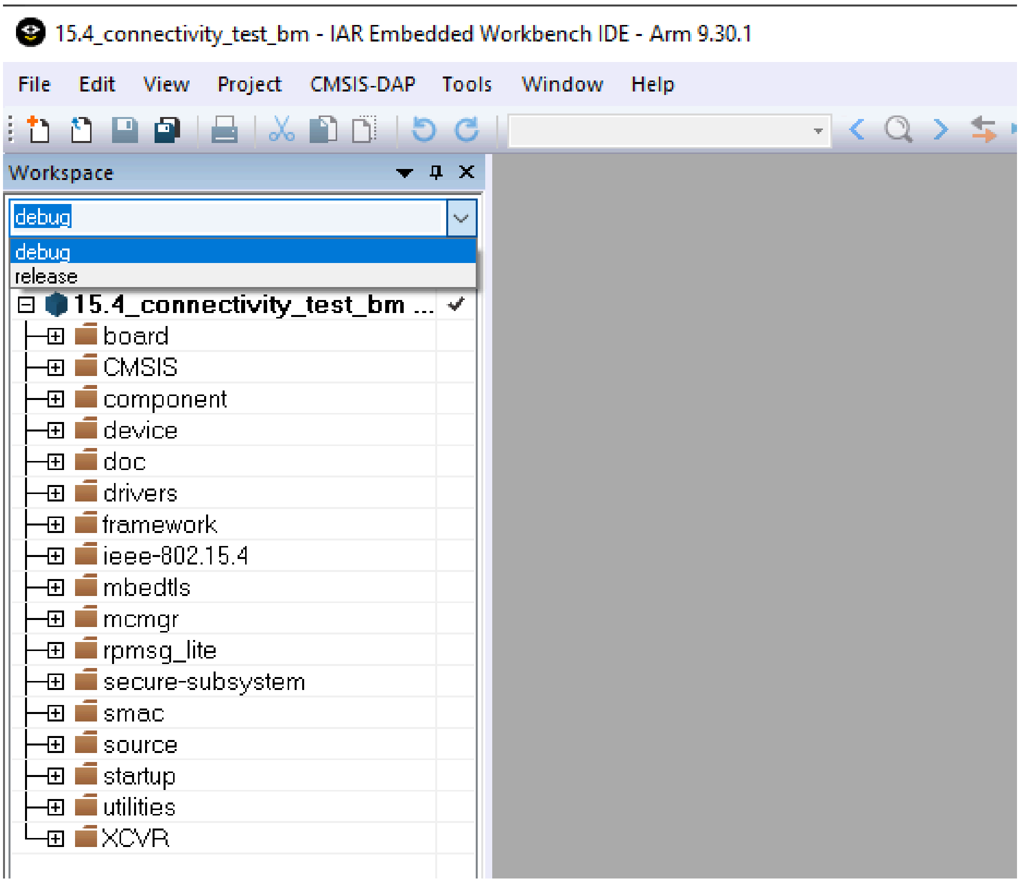

Figure 2. Connectivity Test Application Configuration Selection

4. **Step 4**: Build the project.

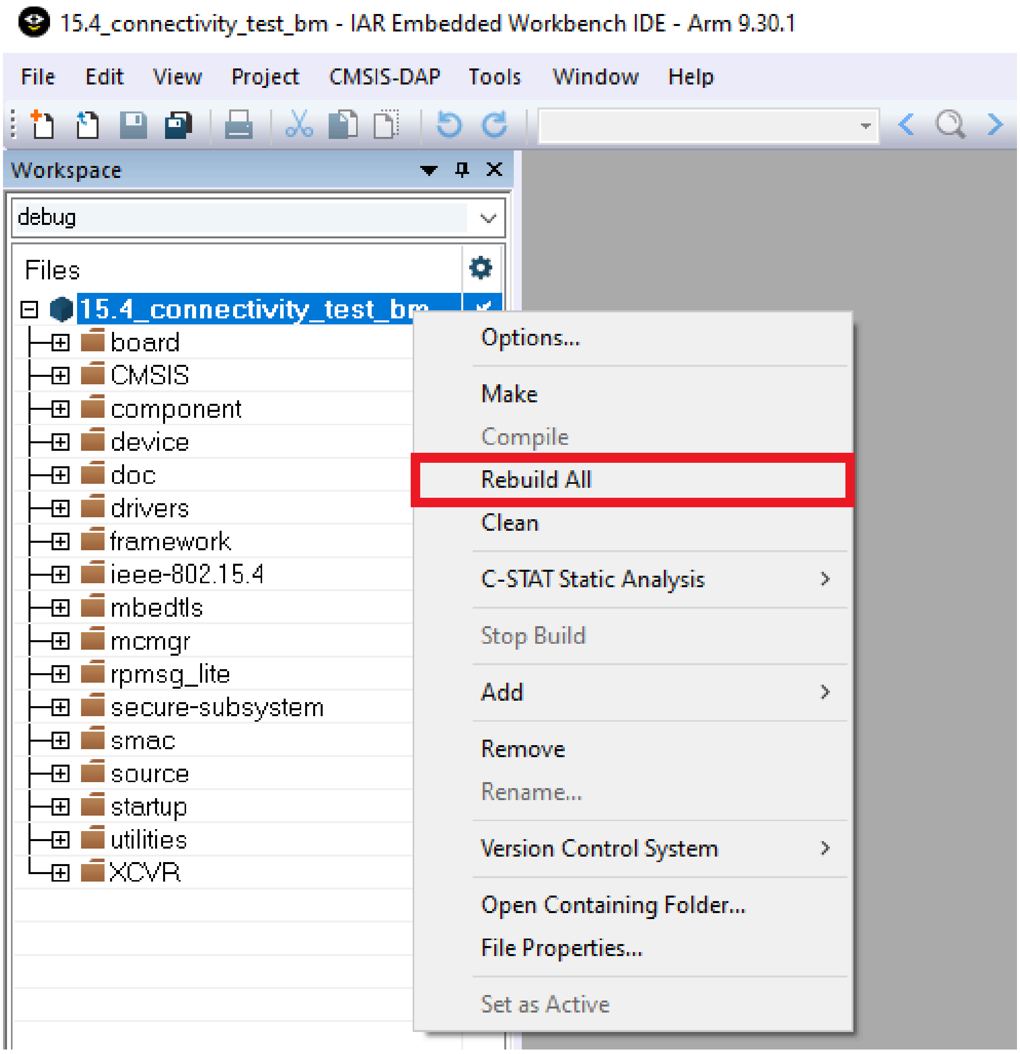

Figure 3. Connectivity Test Application project build

5. **Step 5**: Make the appropriate debugger settings in the project options window:

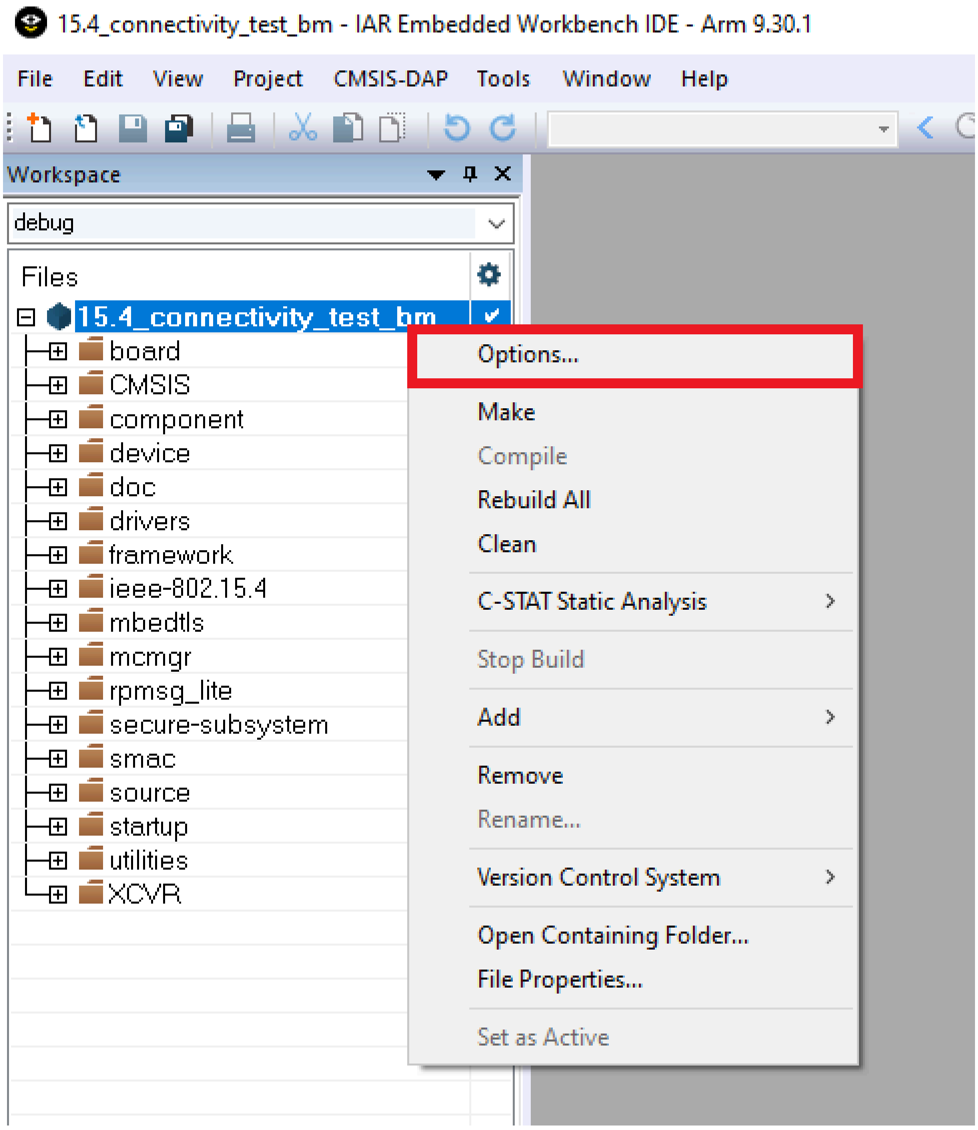

Figure 4. Connectivity Test Application navigate to project options window

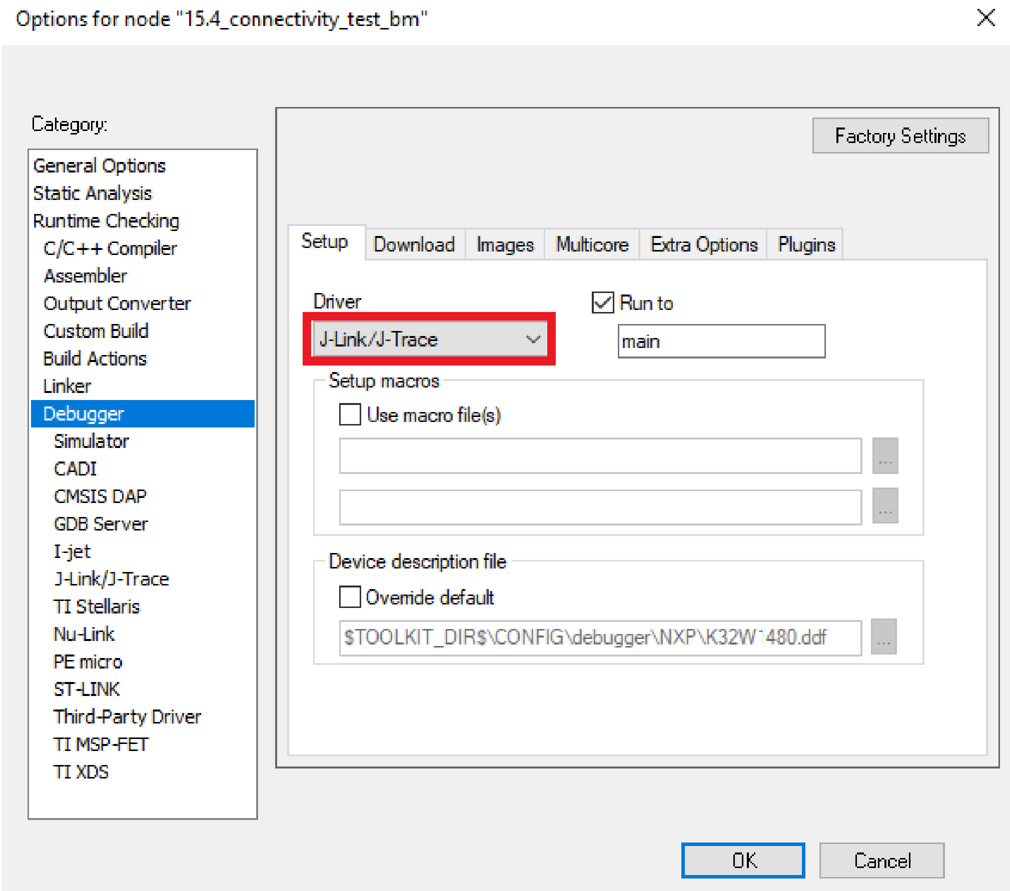

Figure 5. Connectivity Test Application Debugger Settings

6. **Step 6**: Click the "**Download and Debug**" button to flash the executable onto the board.

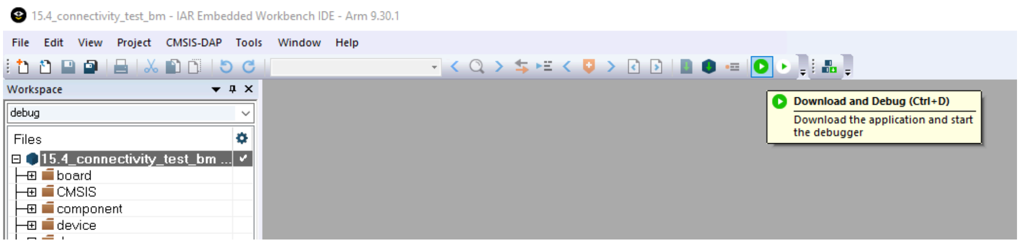

Figure 6. Connectivity Test Application Download and Debug

4 Running the Connectivity Test application

This section demonstrates the basic steps to run a demo application.

Note: *For detailed information about the Connectivity Test Application, refer to the "Connectivity Test Application Command Line Interface User Guide" included in the package (CTACLIUG_Rev0.pdf).*

In order to connect to the board, the Connectivity Test application requires a serial terminal program. This example uses Tera Term for this purpose.

1. **Step 1**: Load the application on the board using IAR Embedded Workbench for Arm by clicking "**Download and Debug**".
2. **Step 2**: After loading the application, check "**Device Manager**" to get the serial port number. This should appear with the prefix "**JLink**".

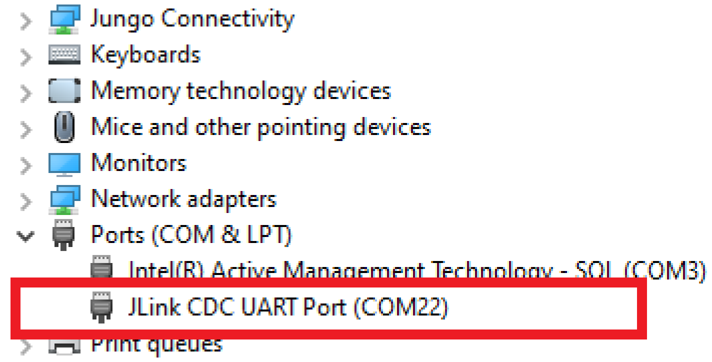

Figure 7. Device Manager serial port lookup

3. **Step 3**: Using the port numbers specified in Device Manager, open a Tera Term instance and connect to the device using the 115200 baud rate. To change the baud rate of the terminal go to the "**Setup** -> **Serial Port**" menu.

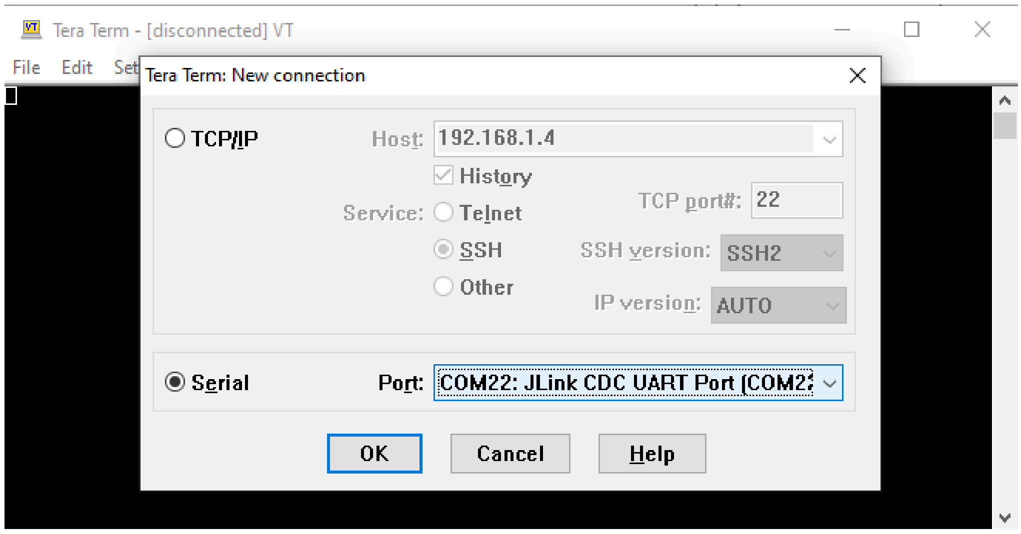

Figure 8. Select JLink serial connection COM port

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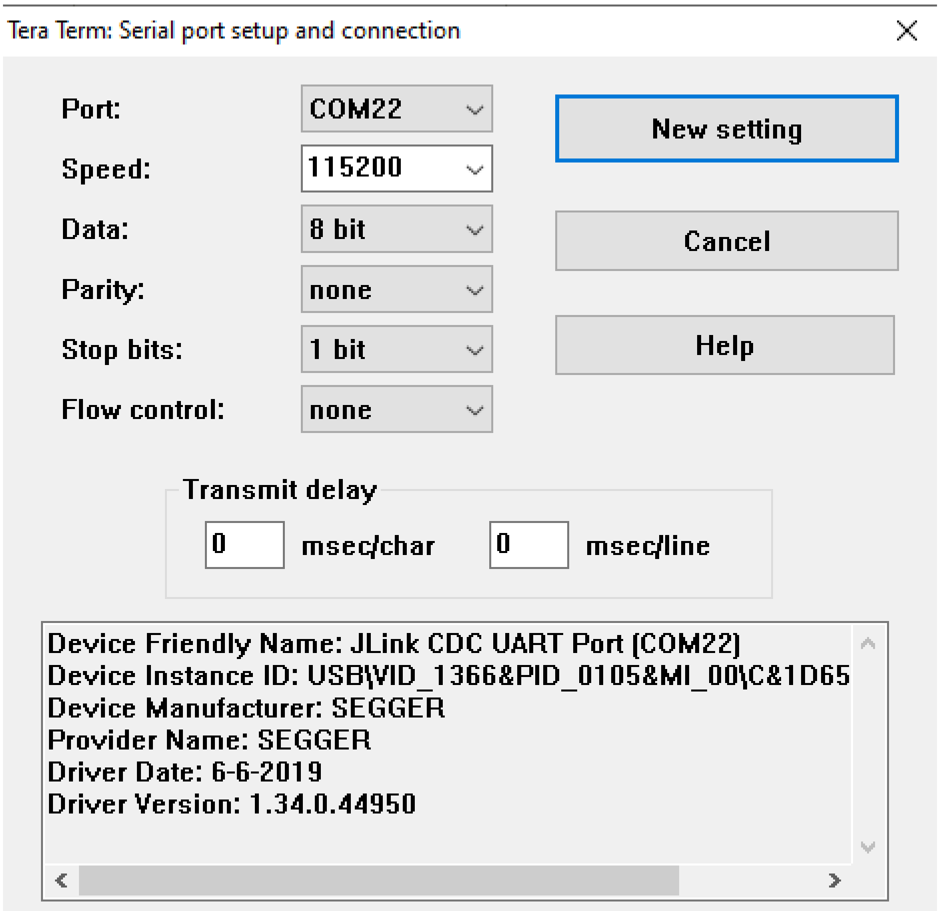

Figure 9. Setting the correct baud rate

4. **Step 4**: Start the application by pressing the “**Enter**” key. Any other key displays the logo screen again.

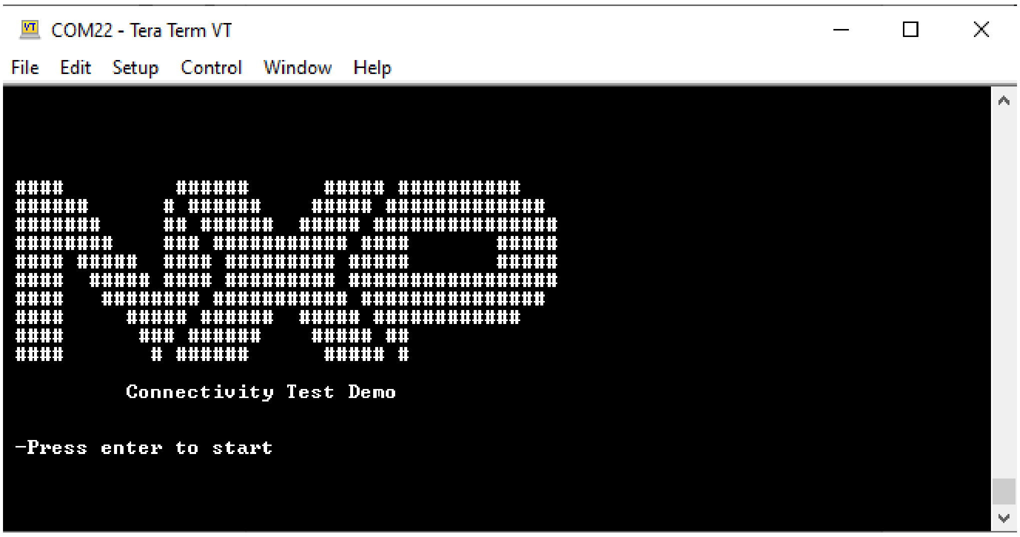

Figure 10. Connectivity Test Application CLI after reset

5. Follow the on-screen instructions to run each test. If a test needs a second platform, follow the steps above to set it up.

5 Note about the source code in the document

Example code shown in this document has the following copyright and BSD-3-Clause license:

6 Revision history

The table below summarizes the revisions to this document.

Table 1. Revision history

Document ID	Release date	Description
UG10204 v.1.0	24 January 2025	Added support for MCXW71 and MCXW72 devices
K32W1CTAGSG v.0.0	27 February 2023	Initial release for K32W1 platform

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